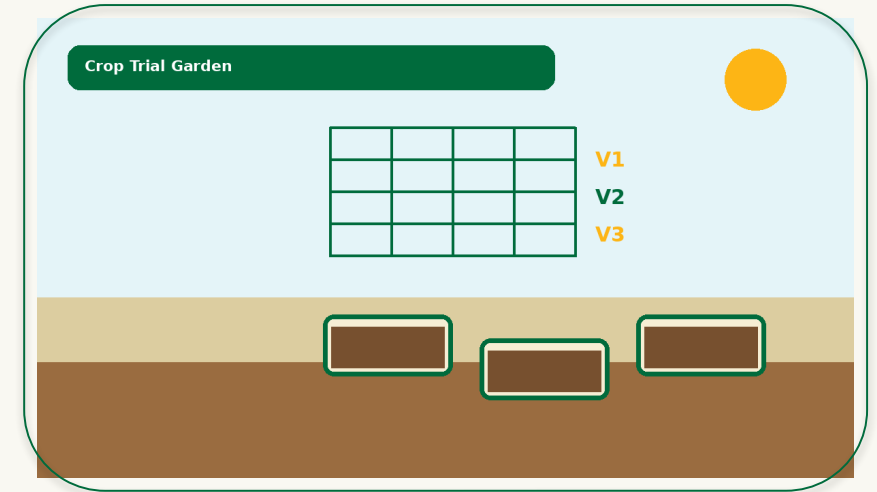


Crop Trial Garden

Designing and Managing Crop Trials

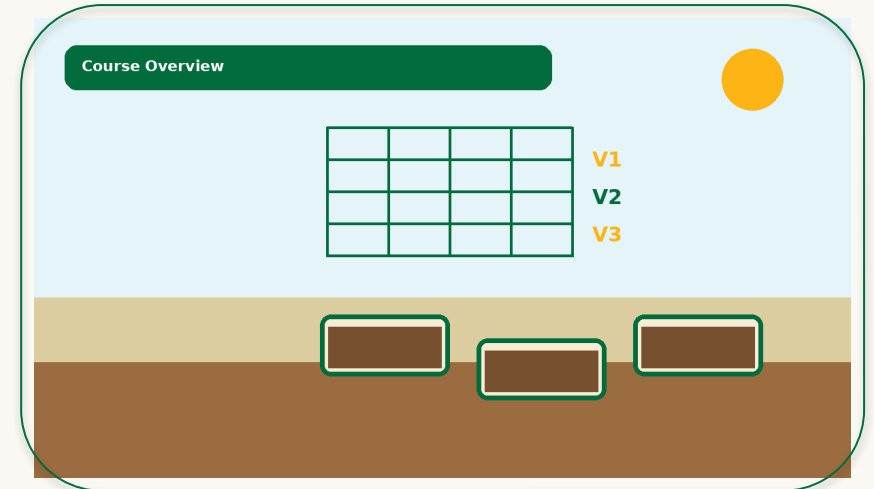
- LIFEWS AgriShine™ GrowMeal™ SSS 1-3 Teacher Training
- Audience: SSS 1-3 Agricultural Science, Biology, Mathematics and technical/STEM educators
- Format: Self-paced online course + supervised field/laboratory practical
- Recommended duration: 4 hours online + 2-3 hours practical
- SSS learning progression: design -> instrument -> model -> evaluate -> optimize -> innovate



Course Overview

Learning outcomes, materials and teacher readiness

- Learning outcomes: explain the system at senior-secondary technical depth; teach all ten modules accurately; supervise design, instrumentation, measurement and experimental work; maintain defensible technical records; guide basic quantitative analysis; distinguish measured evidence from inference; connect results to management and engineering decisions.
- Core materials: teacher guide, system/plot drawings, crop and experiment plans, approved field tools, irrigation equipment, measuring tapes, scales, flow/pressure/moisture instruments where relevant, sensors/data sheets, clean harvest/sample containers, PPE/handwashing supplies and first-aid access.
- Completion expectation: complete all modules, demonstrate one safe technical practical, and submit a production plan, engineering design, experimental protocol or analytical report appropriate to the course.



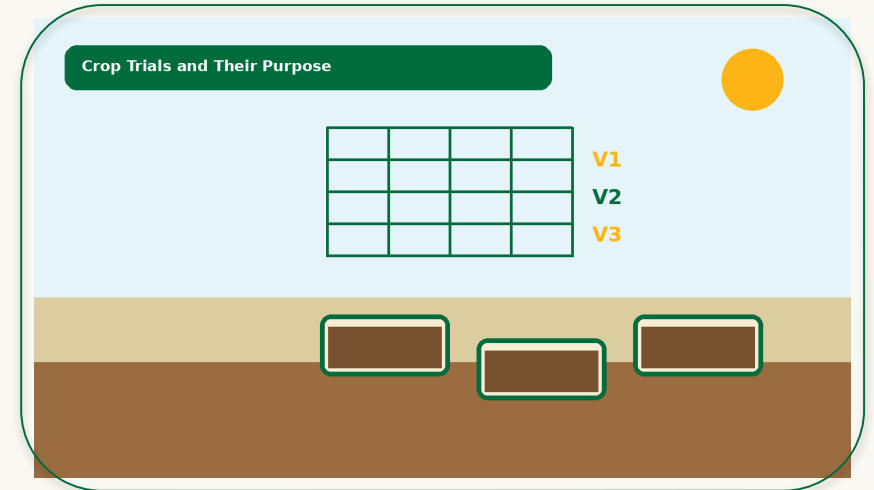
TEACHER PRACTICE

- Training sequence: DEFINE -> DESIGN -> IMPLEMENT -> MEASURE -> ANALYZE -> EVALUATE -> IMPROVE.

Crop Trials and Their Purpose

Teacher technical knowledge, laboratory/field practice and student supervision

- Define a crop trial as a structured comparison designed to evaluate crop varieties or management options under specified conditions.
- State the decision the trial is intended to inform.
- Distinguish demonstration plots from experiments capable of stronger inference.
- Define the target environment and performance criteria before planting.
- Recognize that one school trial provides local evidence, not universal proof.



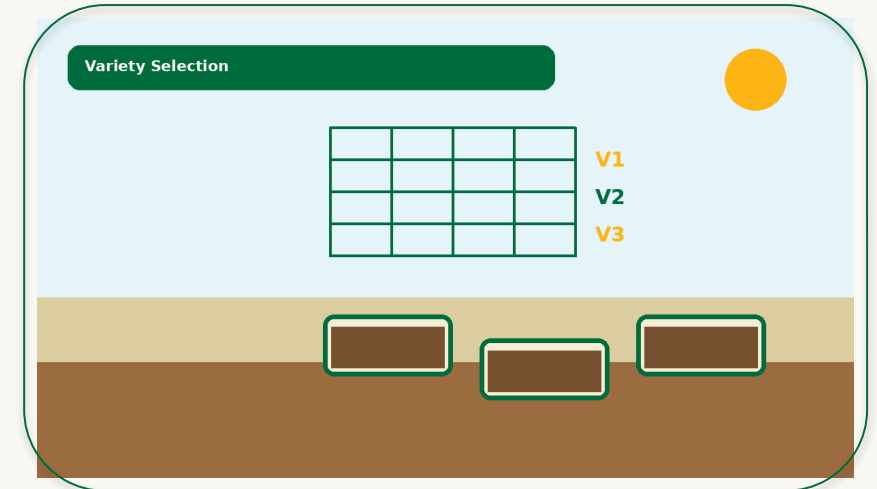
TEACHER PRACTICE

- Teacher task: write the purpose and decision question for a variety trial.

Variety Selection

Teacher technical knowledge, laboratory/field practice and student supervision

- Select varieties with a clear reason for comparison such as maturity, yield, quality or adaptation.
- Use reliable seed sources and record variety identity.
- Choose a manageable number of varieties for available land and replication.
- Avoid confounding variety with different planting dates or management.
- Document relevant known characteristics without prejudging trial results.



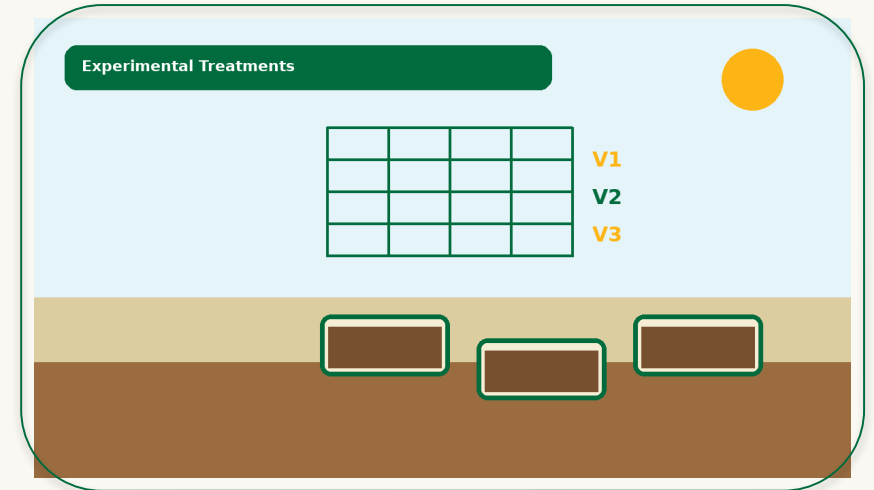
TEACHER PRACTICE

- Prepare a variety-selection table with selection criteria.

Experimental Treatments

Teacher technical knowledge, laboratory/field practice and student supervision

- Treat each variety or management condition as a clearly defined treatment.
- Standardize seed rate, planting method and crop management unless those factors are intentionally tested.
- Label treatments using codes as well as names.
- Record protocol deviations.
- Avoid adding extra differences that make variety effects difficult to interpret.



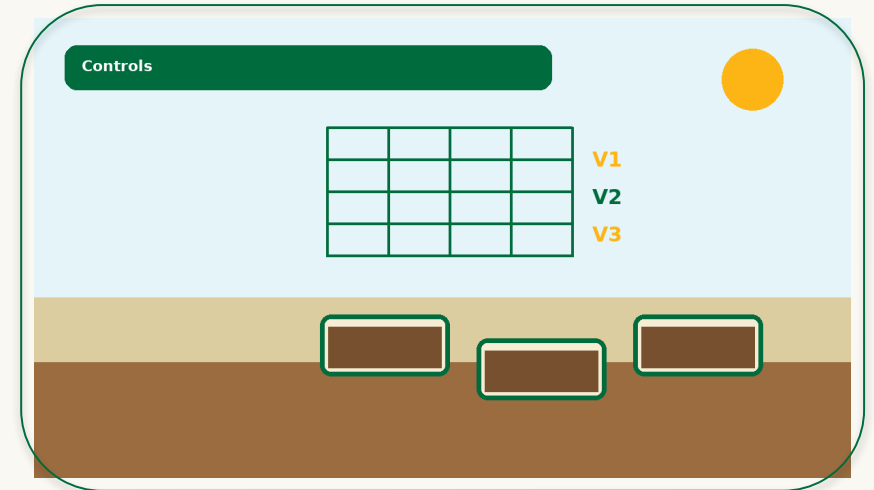
TEACHER PRACTICE

- Write a treatment specification sheet.

Controls

Teacher technical knowledge, laboratory/field practice and student supervision

- Use a standard, commonly grown or reference variety when a control/check is meaningful.
- Manage the control identically to other varieties except for inherent genetic differences.
- Replicate the control like other treatments.
- Explain why the selected check provides a useful benchmark.
- Do not describe the control as 'untreated' when the trial is comparing varieties.



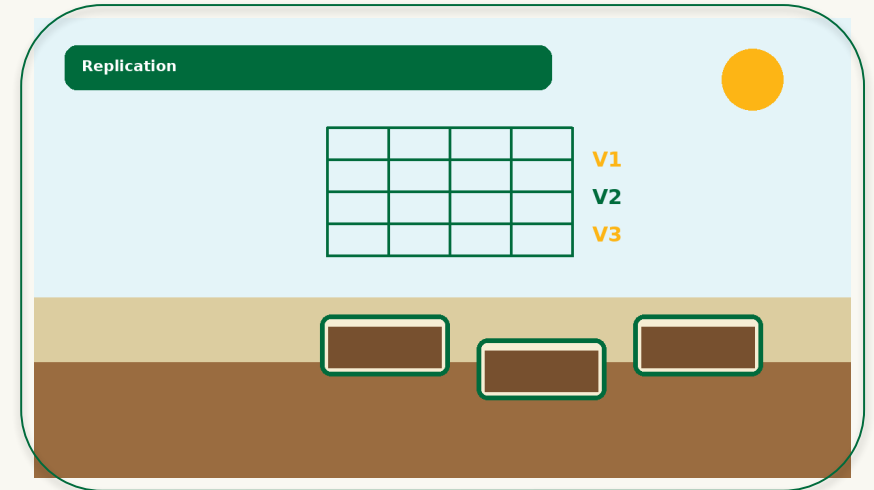
TEACHER PRACTICE

- Select and justify a check variety.

Replication

Teacher technical knowledge, laboratory/field practice and student supervision

- Use repeated plots of each variety to estimate field-to-field variation.
- Distinguish replication from measuring many plants within one plot.
- Choose a feasible replicate number while preserving adequate plot size.
- Maintain each replicate using the same protocol.
- Use replication in the analysis rather than averaging everything prematurely.



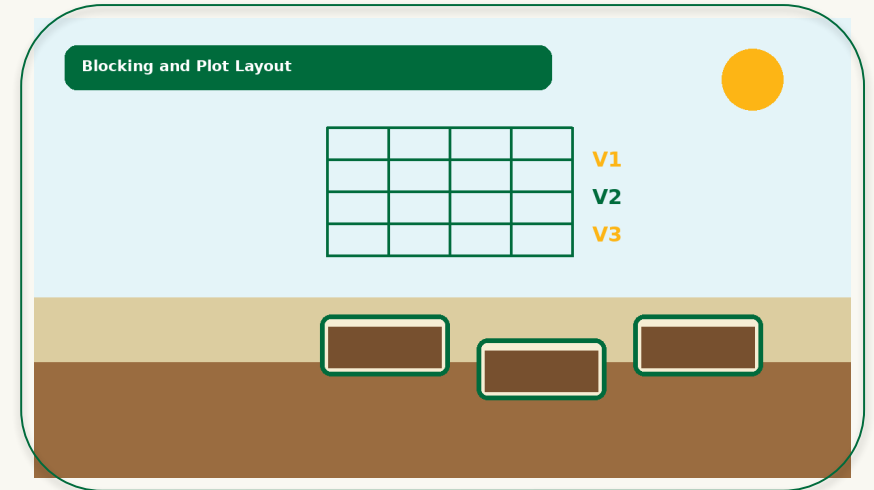
TEACHER PRACTICE

- Design a trial with at least three replicates where feasible.

Blocking and Plot Layout

Teacher technical knowledge, laboratory/field practice and student supervision

- Use blocks to group plots experiencing relatively similar field conditions when a gradient is expected.
- Randomize treatment positions within blocks.
- Define plot dimensions, alleys, borders and harvest area.
- Create a permanent field map with plot IDs.
- Prevent irrigation or access patterns from systematically favoring a variety.



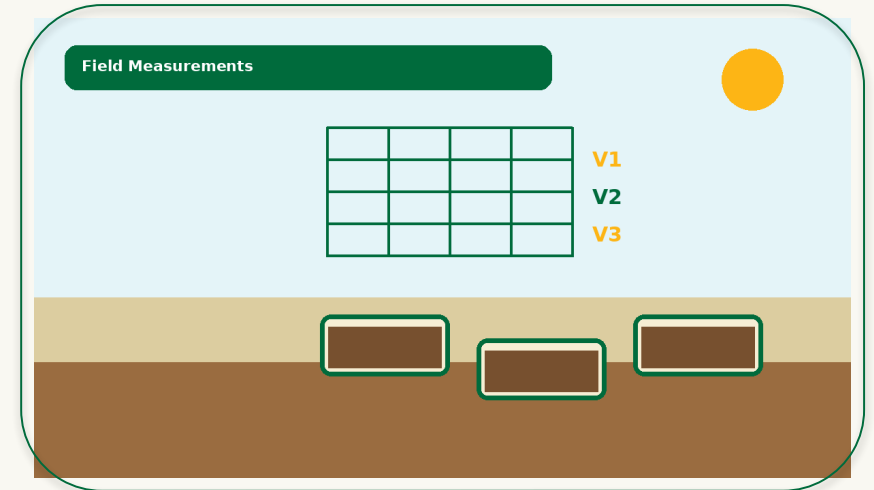
TEACHER PRACTICE

- Prepare a randomized complete block layout for four varieties.

Field Measurements

Teacher technical knowledge, laboratory/field practice and student supervision

- Select traits tied to the trial objective such as emergence, flowering date, height, lodging, pest damage or maturity.
- Write measurement definitions and sampling rules before fieldwork.
- Use the same timing and instruments across varieties.
- Record missing plants and unusual events.
- Train student teams to minimize observer inconsistency.



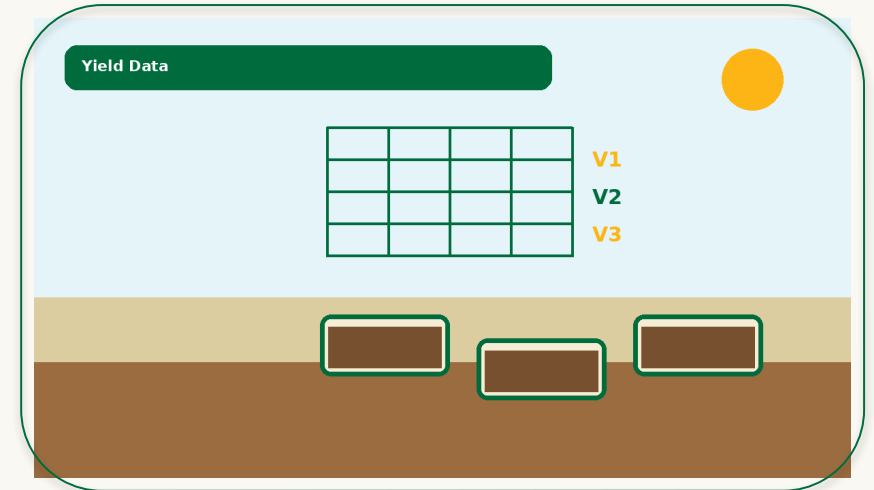
TEACHER PRACTICE

- Develop a field measurement protocol for three traits.

Yield Data

Teacher technical knowledge, laboratory/field practice and student supervision

- Harvest defined plot areas at comparable maturity using consistent criteria.
- Measure yield in standard units and retain plot-level data.
- Record moisture basis or other adjustments if relevant and available.
- Separate marketable and non-marketable yield when quality is part of the objective.
- Check data entry before analysis.



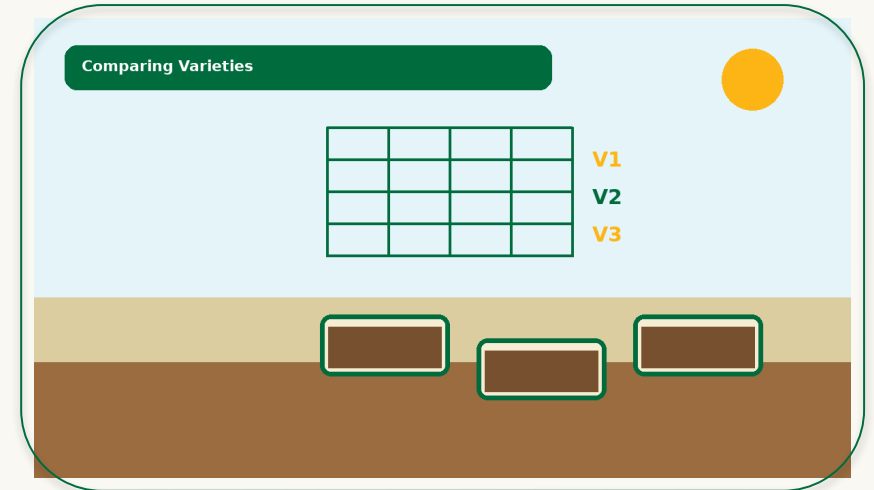
TEACHER PRACTICE

- Create a plot-level yield data sheet.

Comparing Varieties

Teacher technical knowledge, laboratory/field practice and student supervision

- Summarize variety performance using means, ranges or other grade-appropriate statistics.
- Compare multiple relevant traits rather than yield alone.
- Use graphs and tables with units.
- Recognize trade-offs such as early maturity versus total yield.
- Avoid ranking varieties beyond the conditions actually tested.



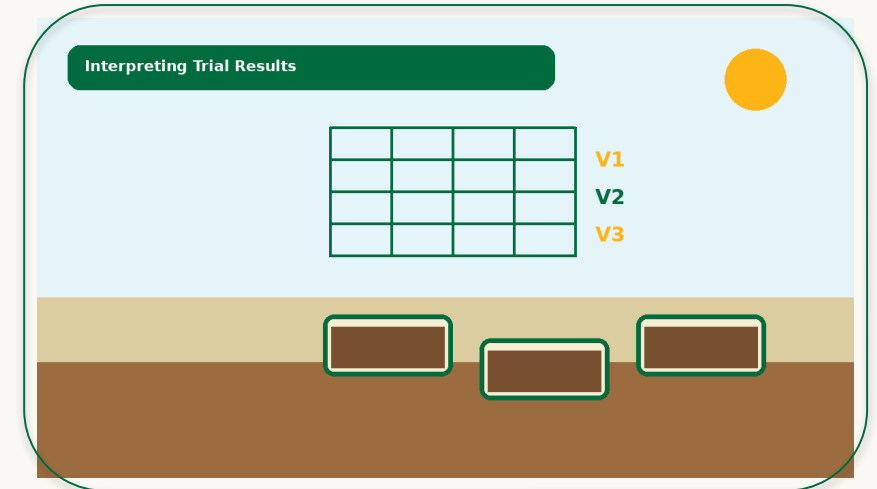
TEACHER PRACTICE

- Prepare a multi-trait variety comparison.

Interpreting Trial Results

Teacher technical knowledge, laboratory/field practice and student supervision

- Relate findings to the original trial purpose and local test conditions.
- Consider variability among replicates and field limitations.
- Distinguish statistical differences from practical importance where analysis allows.
- State which variety appears promising and what evidence supports that judgment.
- Recommend additional seasons/sites before making broad claims.



TEACHER PRACTICE

- Completion task: produce a concise crop-trial interpretation and recommendation.